

Karl Pearson's Correlation Coefficient

Meaning of Correlation: If the change in one variable affects a change in other variable, the variables are said to be correlated.

If the two variables deviate in the same direction, i.e. if the increase (or decrease) in one results in a corresponding increase (or decrease) in other, correlation is said to be direct or positive.

But if they constantly deviate in the opposite directions i.e. if increase (or decrease) in one results in corresponding decrease (or increase) in the other, correlation is said to be diverse (inverse) or negative.

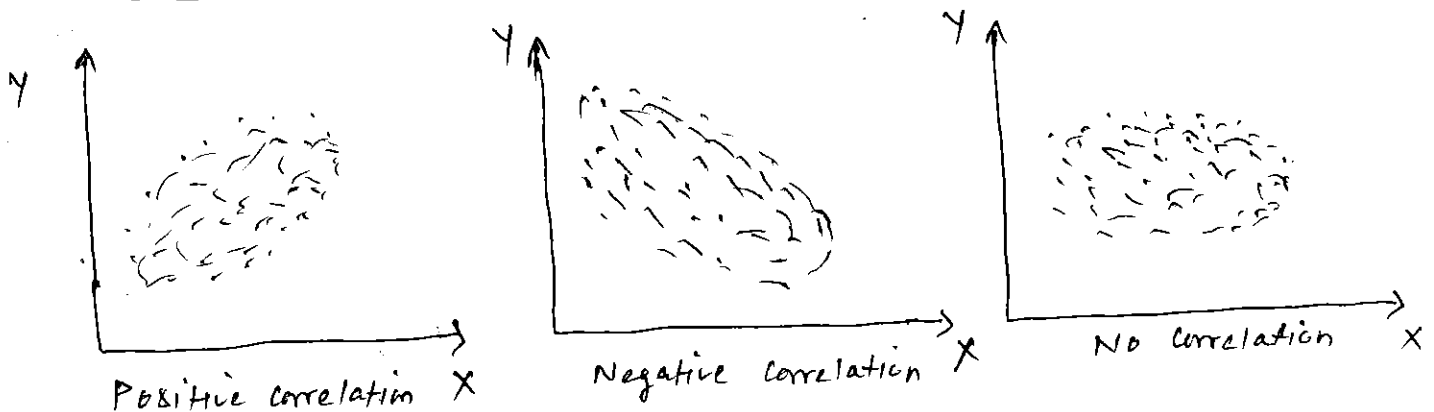
Ex. Positive Correlation: Heights + weights of a group of persons
Income + Expenditure

Negative Correlation: Price + Demand of a commodity
Volume + Pressure of a perfect gas.

Scattered Diagram

Bivariate Data

X	x_1	x_2	...	x_i	...	x_n
Y	y_1	y_2	...	y_i	...	y_n



Karl Pearson's correlation coefficient measures the degree of linear relationship between two variables and is defined as

$$r_{xy} \text{ or } \gamma_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Where $\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

$$\sigma_X^2 = E(X - E(X))^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma_Y^2 = E(Y - E(Y))^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$$

Formula for numerical purpose

$$\begin{aligned}\sigma_{xy} = \text{Cov}(x, y) &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y} \\ &= \frac{1}{n} \sum_{i=1}^n x_i y_i - \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n y_i \right) \\ &= \frac{1}{n^2} \left[n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right]\end{aligned}$$

$$\begin{aligned}\text{Var}(x) = \sigma_x^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \frac{1}{n^2} \left(\sum_{i=1}^n x_i \right)^2 \\ &= \frac{1}{n^2} \left[n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 \right]\end{aligned}$$

$$\begin{aligned}\text{Similarly } \text{Var}(y) = \sigma_y^2 &= \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \frac{1}{n^2} \left[n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2 \right]\end{aligned}$$

$$\begin{aligned}\therefore r_{xy} &= \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2}} \\ &= \frac{\frac{1}{n^2} \left[n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) \right]}{\sqrt{\frac{1}{n^2} \left[n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 \right]} \sqrt{\frac{1}{n^2} \left[n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2 \right]}}\end{aligned}$$

$$r_{xy} = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{\sqrt{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \sqrt{n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2}}$$

→ Suitable for practical purpose
(least approximations required)

Limits of Correlation Coefficient

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sigma_X \cdot \sigma_Y}$$

$$= \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma_X} \right) \left(\frac{y_i - \bar{y}}{\sigma_Y} \right)$$

$$\text{Let } x_i' = \frac{x_i - \bar{x}}{\sigma_X} \quad \text{and} \quad y_i' = \frac{y_i - \bar{y}}{\sigma_Y}$$

$$\text{then } r_{xy} = \frac{1}{n} \sum_{i=1}^n x_i' y_i' , \quad \sum_{i=1}^n x_i'^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma_X^2} = \frac{n \sigma_X^2}{\sigma_X^2} = n$$

$$\sum_{i=1}^n y_i'^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{\sigma_Y^2} = \frac{n \sigma_Y^2}{\sigma_Y^2} = n$$

$$\text{Consider } \frac{1}{n} \sum_{i=1}^n (x_i' + y_i')^2 \geq 0$$

$$= \frac{1}{n} \left[\sum_{i=1}^n x_i'^2 + \sum_{i=1}^n y_i'^2 + 2 \sum_{i=1}^n x_i' y_i' \right] \geq 0$$

$$\Rightarrow \frac{1}{n} [n + n + 2n r_{xy}] \geq 0$$

$$\Rightarrow 1 + 1 + 2 r_{xy} \geq 0$$

$$\Rightarrow r_{xy} \geq -1 \quad \text{--- (X)}$$

$$\text{Again consider } \frac{1}{n} \sum_{i=1}^n (x_i' - y_i')^2 \geq 0$$

$$\Rightarrow \frac{1}{n} \left[\sum_{i=1}^n x_i'^2 + \sum_{i=1}^n y_i'^2 - 2 \sum_{i=1}^n x_i' y_i' \right] \geq 0$$

$$\Rightarrow \frac{1}{n} [n + n - 2n r_{xy}] \geq 0$$

$$\Rightarrow 1 + 1 - 2 r_{xy} \geq 0$$

$$\Rightarrow r_{xy} \leq 1 \quad \text{--- (XX)}$$

Hence from (X) & (XX)

$$\boxed{-1 \leq r_{xy} \leq 1}$$

$$\text{or } |r_{xy}| \leq 1$$

Effect of change of origin + scale on correlation coefficient.

For $\frac{X}{Y}$
$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

Consider the transformations

$$U = \frac{X-A}{h} \quad \text{and} \quad V = \frac{Y-B}{k}$$

Where A, h, B & k are suitably chosen constants.

$$U_i = \frac{x_i - A}{h}, \quad x_i = A + hU_i$$

$$\bar{x} = A + h\bar{u}$$

$$(x_i - \bar{x}) = h(u_i - \bar{u})$$

Similarly $V_i = \frac{y_i - B}{k}, \quad y_i = B + kV_i$

$$\bar{y} = B + k\bar{v}$$

$$(y_i - \bar{y}) = k(v_i - \bar{v})$$

$$\begin{aligned} \text{Cov}(X, Y) &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{h \cdot k}{n} \sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v}) \\ &= h \cdot k \cdot \text{Cov}(U, V) \end{aligned}$$

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{h^2}{n} \sum_{i=1}^n (u_i - \bar{u})^2 = h^2 \text{Var}(U) = h^2 \sigma_u^2$$

$$\Rightarrow \sigma_x = |h| \sigma_u$$

$$\sigma_y^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{k^2}{n} \sum_{i=1}^n (v_i - \bar{v})^2 = k^2 \text{Var}(V) = k^2 \sigma_v^2$$

$$\Rightarrow \sigma_y = |k| \sigma_v$$

Hence
$$r_{xy} = \frac{h \cdot k \cdot \text{Cov}(U, V)}{|h| |k| \sigma_u \sigma_v} = \frac{h \cdot k}{|h| |k|} r_{uv}$$

i.e.
$$r_{xy} = \frac{h \cdot k}{|h| |k|} r_{uv}$$

Explanation: ① If h & k have same sign then $\frac{h \cdot k}{|h| |k|} = +1$ and $r_{xy} = r_{uv}$

If h & k have opposite sign $\frac{h \cdot k}{|h| |k|} = -1$ and $r_{xy} = -r_{uv}$

Correlation coefficient is independent of change of origin and scale in terms of magnitude.

Ex . Two independent variables are uncorrelated.

Proof: If X and Y are independent variables, then

$$\text{Cov}(X, Y) = 0 \Rightarrow r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 0$$

Hence, two independent variables are uncorrelated.

→ But converse is not true. i.e. two uncorrelated variables may not be independent as the following example illustrates:

X	-3	-2	-1	1	2	3	0	→ ΣX	$\bar{X} = \frac{1}{n} \Sigma X = 0$
Y	9	4	1	1	4	9	28	→ ΣY	$\text{Cov}(X, Y) = \frac{1}{n} \Sigma XY - \bar{X} \bar{Y} = 0$
XY	-27	-8	-1	1	8	27	0	→ ΣXY	$r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 0$

X and Y are uncorrelated but Y and X have the relation $Y = X^2$.

⇒ X and Y are not having linear relationship but may have other type of relation.

More Examples

→ (i) $X \sim N(0, 1)$ and $Y = X^2$, Since $X \sim N(0, 1)$, $E(X) = 0 = E(X^3)$
 $\therefore \text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(X^3) - E(X)E(Y) = 0$
 ($\because Y = X^2$)

$$\Rightarrow r_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 0.$$

Hence, X and Y are uncorrelated but not independent ($\because Y = X^2$)

(ii) $X \sim f(x) = \frac{1}{2}$, $-1 \leq x \leq 1$, and $Y = X^2$

Hence, we shall get $E(X) = 0$ and $E(XY) = E(X^3) = 0$,

so that $r_{xy} = 0$.

Principle of Least Square (Fitting of a curve)

x	x_1	x_2	$\dots$	x_n
y	y_1	y_2	$\dots$	y_n

x: independent variable, y: dependent variable

Y: Age of a tree, X: Girth of the tree

$$y = f(x, a_0, a_1, \dots, a_k)$$

Residual or Error Sum of Squares

$$E = \sum_{i=1}^n (y_i - f(x_i, a_0, a_1, \dots, a_k))^2$$

$$\frac{dE}{da_0} = 0 \quad \text{--- (1)}$$

$$\frac{dE}{da_1} = 0 \quad \text{--- (2)}$$

$$\frac{dE}{da_k} = 0 \quad \text{--- (k+1)}$$

Normal Equations

Solve (K+1) normal equations for $a_0, a_1, \dots, a_k$.
Let the solutions be $a_0 = \hat{a}_0, a_1 = \hat{a}_1, \dots, a_k = \hat{a}_k$

$\therefore$ The best fitted function is

$$y = f(x, \hat{a}_0, \hat{a}_1, \dots, \hat{a}_k)$$

Fitting of a straight line (linear eqn)

Let the required linear eqn be y

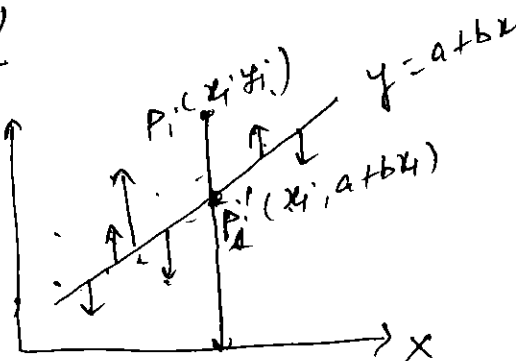
$$y = a + bx$$

Error in estimating the point P_i using the linear eqn $y = a + bx$ is

$$E_i = P_i P_i' = (y_i - (a + bx_i))$$

$$E = \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$

To unknown a, b, we get the normal eqns



$$\frac{dE}{da} = 0 \Rightarrow 2 \sum_{i=1}^n (y_i - a - bx_i)(-1) = 0$$

$$\Rightarrow \sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i \quad \text{--- (i)}$$

$$\frac{dE}{db} = 0 \Rightarrow 2 \sum_{i=1}^n (y_i - a - bx_i)(-x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n y_i x_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \text{--- (ii)}$$

X	Y	X ²	XY
ΣX	ΣY	ΣX ²	ΣXY

Solving Normal Eq^s (i) & (ii)
Let the sol^s be $a = \hat{a}$ and $b = \hat{b}$

∴ Best fitted eqⁿ of S.L. is

$$\underline{y = \hat{a} + \hat{b}x}$$

For any given value of x , we can predict (estimate) y .

Regression Analysis

The term 'regression' literally means "stepping back towards the average"

Defⁿ: "Regression analysis is a mathematical measure of the average relationship between two or more variables in terms of the original units of the data."

In regression analysis there are two types of variables

(i) Dependent variable: The variable whose value is influenced or is to be predicted.

(ii) Independent variable: The variable which influences the values of dependent-variable or is used for prediction.

In regression analysis independent variable is also known as regressor or predictor or explanatory variable while the dependent variable is also known as regressed or explained variable.

Linear Regression

if the variables in a bivariate distribution are related, we will find that the points in the scatter diagram will cluster round some curve called the "curve of regression". If the curve is a straight line, it is called the line of regression and there is said to be linear regression between the variables, otherwise regression is said to be curvilinear.

The line of regression is the line which gives the best estimate to the values of one variable for any specific value of the other variable. Thus the line of regression is the line of "best fit" and is obtained by the principle of least squares.

X	x_1	x_2	-	-	x_n
Y	y_1	y_2	-	-	y_n

Let us suppose that in the bivariate distribution $(x_i, y_i); i=1, 2, \dots, n$, y is the dependent variable and x is the independent variable. Let the line of regression of y on x be

$$Y = a + bx \quad \text{--- (i)}$$

above is the family of straight lines for different arbitrary constants 'a' and 'b'. The problem is to determine 'a' and 'b' so that the line (i) is the line of best fit.

Now using the principle of least square the residual or error sum of squares is written as

$$E = \sum_{i=1}^n P_i H_i^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$

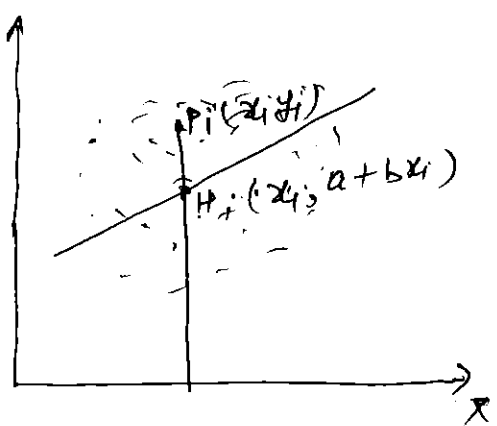
the normal equations are

$$\frac{\partial E}{\partial a} = 0 \Rightarrow 2 \sum_{i=1}^n (y_i - a - bx_i)(-1) = 0$$

$$\Rightarrow \sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i \quad \text{--- (ii)}$$

$$\text{Similarly } \frac{\partial E}{\partial b} = 0 \Rightarrow 2 \sum_{i=1}^n (y_i - a - bx_i)(-x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \text{--- (iii)}$$



Now, we solve the normal equations (ii) + (iii) for the arbitrary constants a and b .

(ii) $\times \sum_{i=1}^n x_i$ + (iii) $\times n$, we get

$$\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right) = na \sum_{i=1}^n x_i + b \left(\sum_{i=1}^n x_i\right)^2 \quad (\text{subtracting})$$

$$- n \sum_{i=1}^n x_i y_i = - na \sum_{i=1}^n x_i - nb \sum_{i=1}^n x_i^2$$

$$\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right) - n \sum_{i=1}^n x_i y_i = b \left(\sum_{i=1}^n x_i\right)^2 - nb \sum_{i=1}^n x_i^2$$

$$\Rightarrow \boxed{b = b_{yx}(\text{slope}) = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2}}$$

b_{yx} is known as the regression coefficient of y on x
 (it gives the rate of change in y with the change in x)
 or for the unit change of x , the y change by the amount b_{yx})

b_{yx} can be further simplified as

$$b_{yx} = \frac{n \sum_{i=1}^n x_i y_i - n\bar{x} \cdot n\bar{y}}{n \sum_{i=1}^n x_i^2 - n^2 \bar{x}^2} = \frac{\frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}}{\frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2}$$

$$= \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\boxed{b_{yx} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}} = \frac{\gamma_{xy} \sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}{\text{Var}(X)}$$

$$= \gamma_{xy} \frac{\sqrt{\text{Var}(Y)}}{\sqrt{\text{Var}(X)}} = \gamma_{xy} \frac{\sigma_y}{\sigma_x}$$

$$\boxed{b_{yx} = \gamma_{xy} \frac{\sigma_y}{\sigma_x}}$$

From normal eqⁿ (ii), we get $\sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i$
 $\Rightarrow n\bar{y} = na + bn\bar{x}$

$$\Rightarrow a = \bar{y} - b\bar{x}$$

$$\Rightarrow a = \bar{y} - b_{yx}\bar{x}$$

Solution of $b = b_{yx}$

Hence from the equation of line of regression proposed in equation (i) we get.

$$y = a + bx$$

$$\Rightarrow y = (\bar{y} - b_{yx}\bar{x}) + b_{yx}x$$

$$\Rightarrow \boxed{y = \bar{y} + b_{yx}(x - \bar{x})} ; \text{ where } b_{yx} = \frac{\text{Cov}(X,Y)}{\text{Var}(X)} = r_{xy} \frac{\sigma_y}{\sigma_x}$$

This is the eqⁿ of regression line of y on x . For any given value of x we can predict the value of y .

$\Rightarrow$ Similarly we can get the equation of regression line of x on y . We start with

$$x = a + by ; \text{ where } a \text{ and } b \text{ are some arbitrary constants.}$$

the normal eqⁿs are

$$\sum_{i=1}^n x_i = na + b \sum_{i=1}^n y_i \quad \text{--- (iv)}$$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n y_i + b \sum_{i=1}^n y_i^2 \quad \text{--- (v)}$$

Solving the normal eqⁿs (iv) and (v) we get.

$$b = b_{xy} = \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2} = \frac{\text{Cov}(X,Y)}{\text{Var}(Y)} = r_{xy} \frac{\sigma_x}{\sigma_y}$$

Hence, $\boxed{b_{xy} = r_{xy} \frac{\sigma_x}{\sigma_y}}$ — regression coefficient of x on y .

From normal eqⁿ (iv), we get $a = \bar{x} - b_{xy}\bar{y}$

Subsequently, the eqⁿ of regression line of x on y is

$$\boxed{x = \bar{x} + b_{xy}(y - \bar{y})} \quad \left\{ \begin{array}{l} \text{From this eqⁿ, we can} \\ \text{predict the value of} \\ x \text{ for any value of } y. \end{array} \right.$$

Remarks: (i) Why there are two eqⁿs of regression lines?

In the eqⁿ of regression line y on x

$$\text{i.e. } y = \bar{y} + b_{yx}(x - \bar{x})$$

y is dependent variable and x is the independent variable, where in the principle of least square (i.e. in error sum of squares), errors are parallel to the y axis.

However, in the eqⁿ of regression line of x on y

$$\text{i.e. } x = \bar{x} + b_{xy}(y - \bar{y}), \quad x \text{ is dependent}$$

variable and y is independent variable where in the error sum of squares, errors are parallel of x -axis. Hence, lines are different

(ii) The two regression coefficients will always have the same sign.

$$\text{i.e. } b_{yx} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = r_{xy} \frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} = r_{xy} \frac{\sigma_x}{\sigma_y}$$

The signs depend upon covariance, that is why both will always have same signs (either both are positive or both are negative).

Properties of Regression Coefficient

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- (i) Correlation coefficient is the geometric mean between the regression coefficients.

$$\text{We know } b_{yx} = r_{xy} \frac{\sigma_y}{\sigma_x} + b_{xy} = r_{xy} \frac{\sigma_x}{\sigma_y}$$

$$\therefore b_{yx} \cdot b_{xy} = r_{xy} \frac{\sigma_y}{\sigma_x} \cdot r_{xy} \frac{\sigma_x}{\sigma_y} = r_{xy}^2$$

$$\Rightarrow r = \pm \sqrt{b_{yx} \cdot b_{xy}}$$

(If regression coefficients are +ve, we will have + sign, if regression coefficients are -ve, we will have - sign of r_{xy})

- (ii) If one of the regression coefficients is greater than unity, the other must be less than unity.

- (1) Let b_{yx} be the greater than unity, then we have to prove $b_{xy} < 1$.

$$\text{Now } b_{yx} > 1 \Rightarrow \frac{1}{b_{yx}} < 1 \quad \text{--- (*)}$$

$$\text{Also we know that } -1 \leq r_{xy} \leq 1 \Rightarrow r_{xy}^2 \leq 1$$

$$\Rightarrow b_{yx} \cdot b_{xy} \leq 1$$

$$\text{Hence } b_{xy} \leq \frac{1}{b_{yx}} < 1 \quad \text{--- (from *)}$$

- (iii) The modulus value of the arithmetic mean of the regression coefficients is not less than the modulus value of the correlation coefficient r_{xy} .

$$\text{i.e. we have to prove that } \left| \frac{1}{2}(b_{yx} + b_{xy}) \right| > |r_{xy}|$$

$$\Rightarrow \left| \frac{1}{2} \left(r_{xy} \frac{\sigma_y}{\sigma_x} + r_{xy} \frac{\sigma_x}{\sigma_y} \right) \right| > |r_{xy}| \Rightarrow \frac{\sigma_y}{\sigma_x} + \frac{\sigma_x}{\sigma_y} > 2 \quad (\because |r| > 0)$$

$$\Rightarrow \sigma_y^2 + \sigma_x^2 - 2\sigma_x\sigma_y > 0 \Rightarrow (\sigma_y - \sigma_x)^2 > 0,$$

which is always true (for real quantity).

- (iv) Effect of change of origin and scale on the regression coefficient.

(Regression coefficients are independent of change of origin but not of scale).

Proof: Let $U = \frac{X-a}{h}$ and $V = \frac{Y-b}{K}$

$\Rightarrow X = a + hU$ and $Y = b + KV$, $\bar{X} = a + h\bar{U}$ and $\bar{Y} = b + K\bar{V}$

Where $a, b, h(>0)$ and $K(>0)$ are constants.

Then $\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = h \cdot K \cdot \frac{1}{n} \sum_{i=1}^n (U_i - \bar{U})(V_i - \bar{V})$

$\therefore = h \cdot K \cdot \text{Cov}(U, V)$

$\text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = h^2 \frac{1}{n} \sum_{i=1}^n (U_i - \bar{U})^2 = h^2 \text{Var}(U)$

Similarly $\text{Var}(Y) = K^2 \text{Var}(V)$

$\therefore b_{YX} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{h \cdot K \cdot \text{Cov}(U, V)}{h^2 \text{Var}(U)} = \frac{K}{h} \frac{\text{Cov}(U, V)}{\text{Var}(U)} = \frac{K}{h} b_{VU}$

$b_{YX} = \frac{K}{h} b_{VU}$

Similarly we can prove that

$b_{XY} = \frac{h}{K} b_{UV}$

Hence, it is proved. it depends only on scales.

Angle Between Two Lines of Regression:

Equations of the lines of regression of Y on X and X on Y are:

$Y = \bar{Y} + r \frac{\sigma_Y}{\sigma_X} (X - \bar{X})$ and $X = \bar{X} + r \frac{\sigma_X}{\sigma_Y} (Y - \bar{Y})$

$\Downarrow$
 $\Rightarrow Y = \bar{Y} + \frac{\sigma_Y}{r \sigma_X} (X - \bar{X})$

Slopes of the lines are $r \cdot \frac{\sigma_Y}{\sigma_X}$ and $\frac{\sigma_Y}{r \sigma_X}$ respectively.

If θ is the acute angle between the two lines of regression, then

$$\tan \theta = \left| \frac{r \frac{\sigma_Y}{\sigma_X} - \frac{\sigma_Y}{r \sigma_X}}{1 + r \frac{\sigma_Y}{\sigma_X} \cdot \frac{\sigma_Y}{r \sigma_X}} \right| = \left| \frac{r^2 - 1}{r} \right| \left(\frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right)$$

$$= \frac{1-r^2}{|r|} \left(\frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right) \quad (\because r^2 \leq 1)$$

$\therefore \theta = \tan^{-1} \left\{ \frac{1-r^2}{|r|} \cdot \left(\frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right) \right\}$

Case (i) (if $r=0$). If $r=0$, $\tan\theta = \infty \Rightarrow \theta = \frac{\pi}{2}$.

Thus the two variables are uncorrelated, the lines of regression become perpendicular to each other.

Case (ii) (if $r=\pm 1$). If $r=\pm 1$, $\tan\theta = 0 \Rightarrow \theta = 0$ or π .

In this case the two lines of regression either coincide or they are parallel to each other. But since both lines of regression pass through the point $(\bar{x}, \bar{y})$, they can not be parallel. Hence in the case of perfect correlation, positive or negative, the two lines of regression coincide.

Remarks: The point of intersection of both regression lines is $(\bar{x}, \bar{y})$.
 $\text{at } \begin{matrix} x = \bar{x} \\ y = \bar{y} \end{matrix} \quad \underline{(x, y) = (\bar{x}, \bar{y})}$

Example: Obtain the equations of two lines of regression for the following data. Also obtain the estimate of x for $y=70$.

$X: 65 \ 66 \ 67 \ 67 \ 68 \ 69 \ 70 \ 72$
 $Y: 67 \ 68 \ 65 \ 68 \ 72 \ 72 \ 69 \ 71$

Solⁿ: Let $U = x - 68$ and $V = y - 69$, then we get
 $\bar{U} = 0, \bar{V} = 0, \sigma_u^2 = 4.5, \sigma_v^2 = 5.5$ $\text{Cov}(u, v) = 3$ and
 $r_{uv} = 0.6$

Since, the correlation coefficient is independent of change of origin, $\therefore r_{xy} = r_{uv} = 0.6$ again if $U = \frac{x-a}{h}, V = \frac{y-b}{k}$, then

~~Again~~ $\bar{x} = a + h\bar{u}, \bar{y} = b + k\bar{v}, \sigma_x = h\sigma_u, \sigma_y = k\sigma_v$

Here $h = k = 1, a = 68, b = 69$

Thus $\bar{x} = 68 + 0 = 68, \bar{y} = 69 + 0 = 69$

$\sigma_x = \sigma_u = \sqrt{4.5} = 2.12$ and $\sigma_y = \sigma_v = \sqrt{5.5} = 2.35$

$\therefore$ Eqⁿ of line of regression of y on x is $y = \bar{y} + b_{yx}(x - \bar{x})$

$$\Rightarrow y = \bar{y} + r_{xy} \frac{\sigma_y}{\sigma_x} (x - \bar{x}) = 69 + 0.6 \frac{2.35}{2.12} (x - 68)$$

$$\Rightarrow y = 0.665x + 23.78$$

Eqⁿ of line of regression of X on Y is

$$\begin{aligned} X &= \bar{X} + b_{xy}(y - \bar{y}) \\ &= \bar{X} + r_{xy} \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \\ &= 68 + 0.6 \frac{2.12}{2.35} (y - 69) \end{aligned}$$

$$\Rightarrow \boxed{X = 0.54y + 30.74}$$

Hence ~~To~~ estimate X for given Y, we use the line of regression of X on Y. $\therefore$ If $Y = 70$,

Estimated value of X is given by

$$\hat{X} = 0.54 \times 70 + 30.74 = 68.54$$

$$\boxed{\hat{X} = 68.54}, \text{ where } \hat{X} \text{ is estimate of } X.$$

Ex. In a partially destroyed laboratory, record of an analysis of correlation data, the following results only are legible:

$\text{Var}(X) = 9$, regression equations: $8X - 10Y + 66 = 0$ and

$$40X - 18Y = 214.$$

What are: (i) the mean values of X and Y (ii) the correlation coefficient between X and Y i.e. r_{xy} and (iii) the standard deviation of Y i.e. σ_y ?

- (i) Since both the lines of regression pass through the point $(\bar{X}, \bar{Y})$, we have: $8\bar{X} - 10\bar{Y} + 66 = 0$ and $40\bar{X} - 18\bar{Y} = 214$. Solving, we get $\bar{X} = 13$, $\bar{Y} = 17$.
- (ii) Let $8X - 10Y + 66 = 0$ and $40X - 18Y = 214$ be the lines of regression of Y on X and X on Y respectively. These equations can be put in the form: $Y = \frac{8}{10}X + \frac{66}{10}$ and $X = \frac{18}{40}Y + \frac{214}{40}$
- $\therefore b_{yx} = \frac{8}{10}$ and $b_{xy} = \frac{18}{40}$
- $\quad \quad \quad = \frac{4}{5} \quad \quad \quad = \frac{9}{20}$

$$\text{Hence } r_{xy}^2 = b_{yx} \cdot b_{xy} = \frac{4}{5} \times \frac{9}{20} = \frac{9}{25}$$

$$\therefore r_{xy} = \pm \frac{3}{5} = \pm 0.6$$

but we take $r_{xy} = 0.6$ as both the regression coefficients are +ve.

(iii) We have $b_{yx} = r_{xy} \frac{\sigma_y}{\sigma_x}$
 $\Rightarrow \frac{4}{5} = \frac{3}{5} \times \frac{\sigma_y}{3}$ gives $\sigma_y = 4$

Remark : If we had assumed that $8x - 10y + 66 = 0$ is the eqⁿ of regression line of x on y and $40x - 18y = 214$ is the eqⁿ of regression line of y on x , then we get respectively

$$8x = 10y - 66 \quad \text{and} \quad 18y = 40x - 214$$

$$x = \frac{10}{8}y - \frac{66}{8} \quad \text{and} \quad y = \frac{40}{18}x - \frac{214}{18}$$

$$\Rightarrow b_{xy} = \frac{10}{8} \quad \text{and} \quad b_{yx} = \frac{40}{18}$$

$$\therefore r_{xy}^2 = b_{xy} \cdot b_{yx} = \frac{10}{8} \times \frac{40}{18} = 2.78 \quad \text{but}$$

$$0 \leq r_{xy}^2 \leq 1. \quad \text{our supposition is wrong.}$$

Ex. Find the most likely price in Mumbai corresponding to the price of Rs 70 at Kolkata from the following:

Average price	Kolkata	Mumbai
	65	67
Standard deviation	2.5	3.5

Correlation coefficient between the prices of commodities in the two cities is 0.8.

Solⁿ: Let the prices (in rupees) in Kolkata and Mumbai be denoted by x and y respectively. Then we are given

$$\bar{x} = 65, \bar{y} = 67, \sigma_x = 2.5, \sigma_y = 3.5, r_{xy} = 0.8$$

We want y for $x = 70$

$\therefore$ The eqⁿ of regression line of y on x is

$$y = \bar{y} + r_{xy} \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$= 67 + 0.8 \frac{3.5}{2.5} (x - 65)$$

$$\text{When } x = 70, \quad y = 67 + 0.8 \frac{3.5}{2.5} (70 - 65) = 72.6$$

Hence, the most likely price in Mumbai corresponding to the price of Rs 70 at Kolkata is Rs. 72.60

Ex. Given that $n=10$, $V = \frac{X-34}{2}$, $W = \frac{Y-45}{3}$,

$$\Sigma V = -6, \Sigma W = 17, \Sigma VW = 37, \Sigma V^2 = 92, \Sigma W^2 = 127$$

- (i) Find r_{xy} (ii) Estimate the value of Y when X is 36.5
(iii) Estimate the value of X when Y is 56.5

(i) The signs of scales are +ve $\therefore r_{xy} = r_{vw}$

$$r_{vw} = \frac{n \Sigma VW - (\Sigma V)(\Sigma W)}{\sqrt{n \Sigma V^2 - (\Sigma V)^2} \cdot \sqrt{n \Sigma W^2 - (\Sigma W)^2}} = r_{xy}$$

Substitute the values and simplify you will get $r_{vw} = r_{xy}$.

(ii) We have to find the eqⁿ of regression line of Y on X

$$i.e. \quad Y = \bar{y} + b_{yx}(X - \bar{x})$$

$$\bar{y} = 45 + 3\bar{W}, \quad \bar{W} = \frac{1}{10} \Sigma W$$

$$\bar{x} = 34 + 2\bar{V}, \quad \bar{V} = \frac{1}{10} \Sigma V$$

$$b_{yx} = \frac{3}{2} b_{wv}, \quad \text{where } b_{wv} = \frac{n \Sigma WV - (\Sigma V)(\Sigma W)}{n \Sigma V^2 - (\Sigma V)^2}$$

Calculate all the terms, you will get the eqⁿ of regression line of Y and X and then estimate Y when X is 36.5.

(iii) We have to find the eqⁿ of regression line of X on Y

$$X = \bar{x} + b_{xy}(Y - \bar{y})$$

$$\bar{x} \text{ \& } \bar{y} \text{ are shown in (ii), now } b_{xy} = \frac{2}{3} b_{vw}$$

$$\text{where } b_{vw} = \frac{n \Sigma VW - (\Sigma V)(\Sigma W)}{n \Sigma W^2 - (\Sigma W)^2}$$

Calculate the terms, you will get the eqⁿ of regression line of X on Y and then estimate X when Y is 56.5.

Ex. Given that $n = 10$, $V = \frac{X-24}{2}$, $W = \frac{Y-64}{4}$
 $\Sigma V = -3$, $\Sigma W = 7$, $\Sigma VW = 23$, $\Sigma V^2 = 55$, $\Sigma W^2 = 43$

Estimate the value of (i) Y when X is 25.0 and (ii) X when Y is 70.0.

Hint: Exactly similar to the previous example. See (i) + (ii).

Ex. Find the coefficient of correlation between X and Y ,
 given that $n = 12$, $\Sigma(X-44) = -5$, $\Sigma(Y-26) = -6$,
 $\Sigma(X-44)^2 = 255$, $\Sigma(Y-26)^2 = 704$ and
 $\Sigma(X-44)(Y-26)/5 = -61.20$.

Let $V = X - 44$ and $W = (Y - 26)$, hence it is
 given $\Sigma V = -5$, $\Sigma W = -6$, $\Sigma V^2 = 255$, $\Sigma W^2 = 704$
 and $\frac{\Sigma VW}{5} = -61.20 \Rightarrow \Sigma VW = -61.20 \times 5$

$$\text{Here } r_{xy} = r_{vw} = \frac{n \Sigma VW - (\Sigma V)(\Sigma W)}{\sqrt{n \Sigma V^2 - (\Sigma V)^2} \sqrt{n \Sigma W^2 - (\Sigma W)^2}}$$

Substitute the given values and simplify,
 you will get r_{xy} .

Let us suppose that a group of n individuals is arranged in order of merit or proficiency in possession of two characteristics A and B. These ranks in the two characteristics will, in general, be different. For example, if we consider the relation between intelligence and beauty, it is not necessary that a beautiful individual is intelligent also. Let $(x_i, y_i); (i = 1, 2, \dots, n)$ be the ranks of the i th individual in two characteristics A and B respectively. Pearsonian coefficient of correlation between the ranks x_i 's and y_i 's is called the rank correlation coefficient between A and B for that group of individuals.

Spearman's Rank Correlation coefficient :

Rank (A)	x_1	x_2	-	-	-	x_n
Rank (B)	y_1	y_2	-	-	-	y_n
$d = (x - y)$	d_1	d_2	-	-	-	d_n

$$d_i = (x_i - y_i)$$

$$P = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Tied Ranks : If some of the individuals receive the same rank in the ranking of merit, they are said to be tied. Let us suppose that m of the individuals say $(k+1)^{th}, (k+2)^{th}, \dots, (k+m)^{th}$ are tied. Then each of these individuals is assigned a common rank, which is arithmetic mean of the ranks $k+1, k+2, \dots, k+m$.

Suppose that there are k such sets of ranks to be tied in the X -series and t such sets of ranks to be tied in the Y -series.

In this rank correlation coefficient is calculated by the formula:

$$P = 1 - \frac{6(\sum d^2 + T_x + T_y)}{n(n^2 - 1)}$$

which is a commonly used formula.

Where $T_x = \frac{1}{12} \sum_{i=1}^k (m_i^3 - m_i)$

and $T_y = \frac{1}{12} \sum_{j=1}^t (m_j'^3 - m_j')$

Limit $-1 \leq P \leq 1$

Ex

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total
Ranks Math (X)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Ranks Physics (Y)	1	10	3	4	5	7	2	6	8	11	15	9	14	12	16	13	
$d = (X - Y)$	0	-8	0	0	0	-1	5	2	1	-1	-4	3	-1	2	-1	3	0
d^2	0	64	0	0	0	1	25	4	1	1	16	9	1	4	1	9	136

Rank correlation coefficient $P = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6 \times 136}{16 \times 255}$
 $= 1 - \frac{1}{5} = 0.8$

Ex Ten competitors in a musical test were ranked by the three Judges A, B and C in the following order:

X. Ranks (A): 1 6 5 10 3 2 4 9 7 8

Y Ranks (B): 3 5 8 4 7 10 2 1 6 9

Z Ranks (C): 6 4 9 8 1 2 3 10 5 7

using rank correlation method, discuss which pair of judges has the nearest approach to common linkings in music.

We get. $P_{xy} = -\frac{7}{33}$, $P_{xz} = \frac{7}{11}$, $P_{yz} = -\frac{49}{165}$, P_{xz} is maximum, A & C has the nearest approach.

Ex. Obtain the rank correlation coefficient to the following data:

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

Calculation for Rank Correlation

	X	Y	R(X) (x)	R(Y) (y)	d = (x - y)	d ²
	68	62	4	5	-1	1
*	64	58	6	7	-1	1
→	75	68	2.5	3.5	-1	1
	50	45	9	10	-1	1
*	64	81	6	1	5	25
	80	60	1	6	-5	25
→	75	68	2.5	3.5	-1	1
	40	48	10	9	1	1
	55	50	8	8	0	0
*	64	70	6	2	4	16
Total					0	72

$$T_x = \frac{3^3 - 2}{12} + \frac{3^3 - 3}{12} = \frac{5}{2}$$

$$T_y = \frac{2^3 - 2}{12} = \frac{6}{12} = \frac{1}{2}$$

$$\therefore P = 1 - \frac{6(\sum d^2 + \frac{5}{2} + \frac{1}{2})}{n(n^2 - 1)}$$

$$= 1 - \frac{6(72 + 3)}{10 \times 99} = \underline{\underline{0.545}}$$

Multiple and Partial Correlation

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When the values of one variable are associated with or influenced by other variable (e.g. age of husband + wife, the height of father + son, the supply + demand of a commodity etc), Karl Pearson's coefficient of correlation can be used as a measure of linear relationship between them. But sometimes there is interrelation between many variables and the value of one variable may be influenced by many others, e.g. the yield of crop per acre, $Y_1(X_1)$ depends upon quality of seed (X_2), fertility of soil (X_3), fertilizer used (X_4), irrigation facilities (X_5), weather conditions (X_6) and so on. When ever we are interested in studying the joint effect of a group of variables upon a variable not included in that group, our study is that of multiple correlation and multiple regression.

Suppose in a trivariate or multivariate distribution we are interested in the relationship between two variables only. There are two alternatives, viz, (i) we consider only those two members of the observed data in which the other members have a specified values or (ii) we may eliminate mathematically the effect of other variables on two variables. The first method has the disadvantage that it limits the size of the data and also it will be applicable to only the data in which the other variates have assigned values. In the second method it may not be possible to eliminate the entire influence of the variates but the linear effect can be easily eliminated. "The correlation and regression between only two variates eliminating the linear effect of other variates in them is called the partial correlation and partial regression".

Yule's Notation: Let us consider a distribution involving three random variables X_1, X_2 and X_3 . Then the equation of the plane of regression of X_1 on X_2 and X_3 is:

$$X_1 = a + b_{12.3} X_2 + b_{13.2} X_3 \quad \text{--- (i)}$$

Without loss of generality we can assume that the variables X_1, X_2 and X_3 have been measured from their respective means, so that

$$E(X_1) = E(X_2) = E(X_3) = 0$$

Hence taking expectation both side of (i) we get $a = 0$

Thus the plane of regression of X_1 on X_2 and X_3 becomes

$$X_1 = b_{12.3} X_2 + b_{13.2} X_3 \quad \text{--- (ii)}$$

The coefficients $b_{12.3}$ and $b_{13.2}$ are known as the partial regression coefficients of X_1 on X_2 and X_1 on X_3 respectively.

$e_{1.23} = b_{12.3}X_2 + b_{13.2}X_3$ is called the estimate of X_1 as given in the plane of regression (ii) and the quantity

$$X_{1.23} = X_1 - b_{12.3}X_2 - b_{13.2}X_3 \equiv X_1 - e_{1.23} \quad \text{--- (iii)}$$

is called the error of estimate or residual.

In the general case of n variables $X_1, X_2, \dots, X_n$, the equation of the plane of regression of X_1 on $X_2, X_3, \dots, X_n$ becomes

$$X_1 = b_{12.34\dots n}X_2 + b_{13.24\dots n}X_3 + \dots + b_{1n.23\dots(n-1)}X_n \quad \text{--- (iv)}$$

The error of estimate or residual is given by:

$$X_{1.23\dots n} = X_1 - [b_{12.34\dots n}X_2 + b_{13.24\dots n}X_3 + \dots + b_{1n.23\dots(n-1)}X_n] \quad \text{--- (v)}$$

The notations used here are due to Yule. The subscripts before the dot(.) are known as the primary subscripts and those after ^{the} dot are called secondary subscripts. The order of regression coefficient is determined by the number of secondary subscripts, i.e. $b_{12.3}, b_{12.34}, \dots, b_{12.34\dots n}$ are the regression coefficients of order 1, 2, $\dots, (n-2)$ respectively.

Thus in general, a regression coefficient with p secondary subscripts will be called a regression coefficient of order ' p '. It may be pointed out that the order in which the secondary subscripts are written is immaterial but the order of the primary subscripts is important, e.g. in $b_{12.34\dots n}$, X_2 is independent while X_1 is dependent variable but in $b_{21.34\dots n}$, X_1 is independent while X_2 is dependent variable.

~~Thus~~ Thus, of the two primary subscripts, former refers to dependent variable and the latter to independent variable.

The order of a residual is also determined by the number of secondary subscripts in it, e.g. $X_{1.23}, X_{1.234}, \dots, X_{1.23\dots n}$ are the residuals of order 2, 3, $\dots, (n-1)$ respectively.

Plane of Regression (For three variables)

The equation of the plane of regression of X_1 on X_2 and X_3 is:

$$X_1 = b_{12.3} X_2 + b_{13.2} X_3 \quad \text{--- (1)}$$

The constants 'b' in (1) are determined by the principle of least squares, i.e. by minimising the sum of the squares of the residuals, viz,

$$S = \sum X_{1.23}^2 = \sum (X_1 - b_{12.3} X_2 - b_{13.2} X_3)^2$$

The normal equations for estimating $b_{12.3}$ and $b_{13.2}$ are:

$$\frac{\partial S}{\partial b_{12.3}} = 0 \Rightarrow -2 \sum X_2 (X_1 - b_{12.3} X_2 - b_{13.2} X_3) = 0 \quad \text{--- (2)}$$

$$\frac{\partial S}{\partial b_{13.2}} = 0 \Rightarrow -2 \sum X_3 (X_1 - b_{12.3} X_2 - b_{13.2} X_3) = 0$$

$$\Rightarrow \begin{cases} \sum X_1 X_2 - b_{12.3} \sum X_2^2 - b_{13.2} \sum X_2 X_3 = 0 \\ \sum X_1 X_3 - b_{12.3} \sum X_2 X_3 - b_{13.2} \sum X_3^2 = 0 \end{cases} \quad \text{--- (3)}$$

Since X_i 's are measured from their respective means, we have

$$\sigma_i^2 = \frac{1}{N} \sum X_i^2, \quad \text{Cov}(X_i, X_j) = \frac{1}{N} \sum X_i X_j$$

$$\text{and } r_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sigma_i \sigma_j} = \frac{\sum X_i X_j}{N \sigma_i \sigma_j}$$

N : denotes number of observations

X_1	X_{11}	X_{12}	$\dots$	X_{1N}
X_2	X_{21}	X_{22}	$\dots$	X_{2N}
X_3	X_{31}	X_{32}	$\dots$	X_{3N}

Hence, from (3), we get

$$\begin{cases} r_{12} \sigma_1 \sigma_2 - b_{12.3} \sigma_2^2 - b_{13.2} r_{23} \sigma_2 \sigma_3 = 0 \\ r_{13} \sigma_1 \sigma_3 - b_{12.3} r_{23} \sigma_2 \sigma_3 - b_{13.2} \sigma_3^2 = 0 \end{cases} \quad \text{--- (4)}$$

Solving the equations (4) for $b_{12.3}$ and $b_{13.2}$, we get

$$b_{12.3} = \frac{\begin{vmatrix} r_{12} \sigma_1 & r_{23} \sigma_3 \\ r_{13} \sigma_1 & \sigma_3 \end{vmatrix}}{\begin{vmatrix} \sigma_2 & r_{23} \sigma_3 \\ r_{23} \sigma_2 & \sigma_3 \end{vmatrix}} = \frac{\sigma_1}{\sigma_2} \frac{\begin{vmatrix} r_{12} & r_{23} \\ r_{13} & 1 \end{vmatrix}}{\begin{vmatrix} 1 & r_{23} \\ r_{23} & 1 \end{vmatrix}} \quad \text{--- (5)}$$

Similarly, we will get

$$b_{13.2} = \frac{\sigma_1}{\sigma_3} \frac{\begin{vmatrix} 1 & r_{12} \\ r_{23} & r_{13} \end{vmatrix}}{\begin{vmatrix} 1 & r_{23} \\ r_{23} & 1 \end{vmatrix}} \quad \text{--- (6)}$$

if we write $W = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}$ --- (7)

And w_{ij} is the cofactor of the element in the i th row and j th column of W , we have from (5) & (6)

$$b_{12.3} = -\frac{\sigma_1}{\sigma_2} \frac{w_{12}}{w_{11}} \quad \text{and} \quad b_{13.2} = -\frac{\sigma_1}{\sigma_3} \frac{w_{13}}{w_{11}} \quad \text{--- (8)}$$

Substituting these values in (1), we get the required eqⁿ of the plane of regression of X_1 on X_2 and X_3 as:

$$X_1 = -\frac{\sigma_1}{\sigma_2} \frac{w_{12}}{w_{11}} X_2 - \frac{\sigma_1}{\sigma_3} \frac{w_{13}}{w_{11}} X_3$$

$$\Rightarrow \boxed{\frac{X_1}{\sigma_1} w_{11} + \frac{X_2}{\sigma_2} w_{12} + \frac{X_3}{\sigma_3} w_{13} = 0} \quad \text{--- (9)}$$

Multiple Correlation Coefficient:

In a tri-variate distribution in which each of the variables X_1, X_2 and X_3 has N observations, the multiple correlation coefficient of X_1 on X_2 and X_3 , usually denoted by $R_{1.23}$, is the simple correlation coefficient between X_1 and the joint effect of X_2 and X_3 on X_1 .

In other words $R_{1.23}$ is the correlation coefficient between X_1 and its estimated value as given by the plane of regression of X_1 on X_2 and X_3 , viz,

$$e_{1.23} = b_{12.3} X_2 + b_{13.2} X_3$$

$$\text{we have } X_{1.23} = X_1 - b_{12.3} X_2 - b_{13.2} X_3 = X_1 - e_{1.23} \Rightarrow e_{1.23} = X_1 - X_{1.23}$$

$$\text{Def def}^n \quad R_{1.23} = \frac{\text{Cov}(X_1, e_{1.23})}{\sqrt{V(X_1)} \sqrt{V(e_{1.23})}}$$

Finally we get

$$R_{1.23}^2 = \frac{\gamma_{12}^2 + \gamma_{13}^2 - 2\gamma_{12}\gamma_{13}\gamma_{23}}{(1 - \gamma_{23}^2)}$$

Similarly we can have

$$R_{2.13}^2 = \frac{\gamma_{21}^2 + \gamma_{23}^2 - 2\gamma_{21}\gamma_{23}\gamma_{13}}{(1 - \gamma_{13}^2)}$$

$$\text{and } R_{3.12}^2 = \frac{\gamma_{31}^2 + \gamma_{32}^2 - 2\gamma_{31}\gamma_{32}\gamma_{21}}{(1 - \gamma_{12}^2)}$$

Remarks : (i) Since $\text{Cov}(X_1, e_{1.23}) = V(e_{1.23}) > 0$

$$\Rightarrow 0 \leq R_{1.23} \leq 1$$

(ii) $R_{1.23}$ is not less than any total correlation coefficient
i.e. $R_{1.23} \geq \gamma_{12}, \gamma_{13}, \gamma_{23}$

Partial Correlation Coefficient :

Sometimes the correlation between two variables X_1 and X_2 may be partly due to the correlation ~~between~~ of a third variable X_3 with both X_1 and X_2 . In such a situation, one may want to know what the correlation between X_1 and X_2 would be if the effect of X_3 on each of X_1 and X_2 were eliminated. This correlation is called the partial correlation and the correlation coefficient between X_1 and X_2 after the linear effect of X_3 on each of them has been eliminated is called the partial correlation coefficient.

The residual $X_{1.3} = X_1 - b_{13}X_3$ may be regarded as the part of the variable X_1 which remains after the linear effect has been eliminated. Similarly, the residual $X_{2.3} = X_2 - b_{23}X_3$ may be interpreted. Thus the partial correlation coefficient between X_1 and X_2 , usually denoted by $\gamma_{12.3}$ is given by:

$$r_{12.3} = \frac{\text{Cov}(X_{1.3}, X_{2.3})}{\sqrt{\text{Var}(X_{1.3})} \sqrt{\text{Var}(X_{2.3})}}$$

$$= \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}}$$

Similarly we can have

$$r_{13.2} = \frac{r_{13} - r_{12}r_{32}}{\sqrt{(1-r_{12}^2)(1-r_{32}^2)}}$$

and $r_{23.1} = \frac{r_{23} - r_{21}r_{31}}{\sqrt{(1-r_{21}^2)(1-r_{31}^2)}}$

Ex: From the data relating to the yield of dry bark (X_1), height (X_2) and girth (X_3) for 18 cinchona plants, the following correlation coefficients were obtained:

$$r_{12} = 0.77, \quad r_{13} = 0.72 \quad \text{and} \quad r_{23} = 0.52$$

Find $r_{12.3}$ and $R_{1.23}$

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}} = \frac{0.77 - 0.72 \times 0.52}{\sqrt{(1-(0.72)^2)(1-(0.52)^2)}}$$

$$= 0.62$$

$$R_{1.23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}$$

$$= \frac{(0.77)^2 + (0.72)^2 - 2 \times 0.77 \times 0.72 \times 0.52}{(1 - (0.52)^2)}$$

$$= 0.7334$$

$$R_{1.23} = +0.8564 \quad (\text{Since multiple correlation coefficient is non-negative})$$